Maximilian Dittgen

myd4@cornell.edu | 917.971.5428

RESEARCH INTERESTS

ICTD/HCI4D; Digital Sovereignty for the Global Majority; Human-AI Collaboration; AI Cultural Alignment

EDUCATION

Cornell University — Ithaca, NY
M.Eng. Computer Science, GPA: 3.85 — May 2026
Project: Value-Aligned LLM Writing Assistants for Improving Conversational Outcomes of Online Discussions

Cornell University — Ithaca, NY
B.S. Computer Science, GPA: 3.97 — May 2025
Magna Cum Laude, Minor in Asian American Studies

Boston University — Boston, MA
Undergraduate coursework (transferred), GPA: 4.00 — 2021 – 2022

RESEARCH EXPERIENCE

Research Assistant, Cornell Computing and Information Science — 2024 – Present
Advisor: Aditya Vashistha

- Investigated cultural alignment and utility of LLM writing assistants for guiding constructive online deliberation on divisive topics in non-Western global contexts
- Designed a systematic value-alignment evaluation framework for 192 literature-derived LLM prompting configurations, measuring how well different prompting strategies preserved users' values and stances during Human-AI co-writing activities
- Led end-to-end technical implementation of value-aligned LLM prompting strategies and evaluation framework, including code development, compute provisioning, and data analysis
- Contributed to power analysis and study design for two user studies of LLM writing assistance with 1,028 participants across India, Egypt, and the United States; built the UI and data collection interfaces (HTML/JS)

This work resulted in one co-authored publication (CSCW 2025) and one manuscript under review (CHI 2027).

Research Assistant, Cornell Urban Technology Lab — 2025
Advisor: H. Oliver Gao

- Designed a multi-objective routing algorithm to find Pareto-optimal paths balancing travel time and pollution exposure; modeled health impacts of route choice on NYC commuter disease incidence
- Integrated LLM-based agents into a traffic microsimulation framework to predict realistic commuter route choice as part of prototype navigation app that reduces exposure to traffic-related pollutants

This work resulted in one manuscript under review (Transportation Research Part D).

Research Assistant, Boston University Performance and Energy-Aware Computing Lab — 2022
Advisor: Ayse Coskun

- Benchmarked performance and power consumption of MicroFaaS, a distributed computing cluster of bare-metal microcontrollers 5.6x more energy-efficient than conventional cloud servers
- Debugged the cluster and optimized its kernel configuration to reduce boot time and increase job execution speed

PUBLICATIONS

1. Farhana Shahid, **Maximilian Dittgen**, Stella Zhang, and Aditya Vashistha. Value-Aligned LLMs Improve Conversational Outcomes on Divisive Topics. *Under review at the ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*.
2. Cafer Avci, Zhaoyao Bao, **Maximilian Dittgen**, Danni Lu, Alireza Yazdiani, Mohammad Tayarani, and H. Oliver Gao. Shortest Path or Not? Modeling the Trade-off Between Travel Time and Pollution. *Under review at Transportation Research Part D.*
3. Farhana Shahid, **Maximilian Dittgen**, Mor Naaman, and Aditya Vashistha. Examining Human-AI Collaboration for Co-Writing Constructive Comments Online. In Proc. ACM Hum.-Comput. Interact. Vol. 9, No. 7, Article 410 (CSCW 2025).

SELECTED WRITING

Evaluating Anticolonial Resistance Methods Against the Digital Platform Metropole 2026
Seminar paper for INFO 6210: Information, Technology, and Society, taught by Daniel Susser and Helen Nissenbaum

Ethical Considerations in the Development of Low-Resource African Language LLMs 2024
Course paper for INFO 4505: Computing for Global Development, taught by Aditya Vashistha

INDUSTRY EXPERIENCE

Software Engineer, Apple 2026 – Present
- Developing compute control plane for pre-release private cloud platform

Software Engineering Intern, Amazon Web Services Fall 2025
- Designed and authored a Linux kernel RFC and patch series optimizing ARM interrupt virtualization in KVM
- Led the series through upstream maintainer review, with three patches accepted into mainline Linux

Software Engineering Intern, Capital One Summers 2024, 2025
- Built data pipelines and APIs for Capital One's travel and Zelle payments platforms

Software Engineering Intern, Exostellar Summer 2023
Cornell faculty spinout, acquired by Qualcomm
- Designed and implemented optimization techniques for VM live migration; work acknowledged in SoCC 2024 publication

TEACHING EXPERIENCE

Graduate Teaching Assistant, Cornell University Spring 2026
- Coordinated course operations between professor and team of 30 undergraduate TAs for CS 4300: Language and Information. Held office hours, organized grading sessions, and administered exams on text mining and information retrieval
- Designed and taught a NumPy workshop for students across upper-division computer science courses

Teaching Assistant, Cornell University 2024 – 2025
- Undergraduate TA for CS 4300: Language and Information, and CS 4852: Markets II (Network Design)

HONORS & AWARDS

Outstanding Graduate Teaching Assistant Award, Cornell University 2026
Woods Engineering Travel Grant, Cornell University 2024
Serve in Place Grant, Cornell Einaudi Center for International Studies 2023
Dean's List, Cornell University (all semesters) 2022 – 2025
Presidential Scholarship, Boston University 2021